Module 3: Linear Programming
Lab Problem Set

Required Problems: 1-3
(A partial template for these 3 problems is available on Canvas)
Additional optional problems: 4-5
Problems labelled with an asterisk (*) are more challenging

Submit an Excel workbook with a separate worksheet for each problem (and additional labelled sheets as necessary for the sensitivity reports). Write your answers to the sub-questions of each problem on the corresponding worksheets. Please label your work clearly.

Problem 1

STL Chocolatier is a leading artisanal chocolate company. They have recently decided to redesign the bulk orders that they create and ship internationally. Broadly speaking there are three types of chocolate boxes they manufacture and sell: Value, Regular, and Deluxe. Demand for each of the three types of boxes is robust, and STL wants to find out how many boxes of each kind it should create for its next bulk delivery, with the goal of maximizing profit. The three types have distinct characteristics in terms of chocolates per box, time required to complete a box, and total weight of each box. We assume the profit per box is fixed, and only depends on its type. The following table describes these quantities:

	Weight/piece (kg)	Profit/box ($)	Pieces/box	Labor time/box (minutes)
Value	0.09	0.60	8	3
Regular	0.12	1.10	4	4
Deluxe	0.20	3.00	6	12

The total allowable weight capacity for a typical bulk order is 350 kilograms and the total labor time available is 36 hours.

a. Formulate and solve this problem using Excel's Solver. (Make sure you generate the sensitivity report associated with the optimal solution). What is the optimal solution? What is the optimal objective function value?

Answer the following additional questions <u>based on the information contained in that sensitivity report</u>:

b. Which constraints are active (i.e., "binding") at the optimal solution?

c. Why is the shadow price of the Weight constraint equal to Zero?

d. Suppose the Chocolatier is considering expanding its working hours. Through labor negotiations, they have determined that they can extend the available amount of labor to 39 hours from 36 hours, with a total associated additional cost of $40. Should they take advantage of this?

e. *Are shadow prices always zero or positive (in any linear programming problem, not just this one)?

Problem 2

Clayton Brewery is a small local brewery whose ale and beer are always in demand but whose production is limited by certain raw materials that are in short supply. (Ale is a type of alcoholic beverage, similar to but heavier than beer.) The scarce materials are corn, hops, and barley malt (all of special high quality). A barrel of ale calls for different amounts of ingredients, compared to a barrel of beer, as follows:

Raw material	Raw Material Requirements	
	Ale (per barrel)	Beer (per barrel)
Corn	6 pounds	14 pounds
Hops	5 ounces	4 ounces
Malt	35 pounds	20 pounds

For each day in the near future, Clayton's Brewery will have the following quantities of raw materials available to them: 480 pounds of corn, 175 ounces of hops, and 1,190 pounds of malt. They also estimate that the profit is $15 per barrel of ale and $25 per barrel of beer. Dobb's Brewery believes that it can sell all the beverages it produces.

Let A and B be the daily production level of ale and beer (in barrels) at Clayton's Brewery. The following linear optimization model will determine the best production plan:

Maximize $15A + 25B$ (profit)

Subject to:

(Corn Resource) $6A + 14B \leq 480$

(Hops Resource) $5A + 4B \leq 175$

(Malt Resource) $35A + 20B \leq 1{,}190$

(Nonnegativity) $A, B \geq 0$

a. Use Excel's Solver to find an optimal solution and the corresponding optimal profit.

b. Generate the Sensitivity Report.

c. Which constraints are active (i.e., "binding")? Which are not?

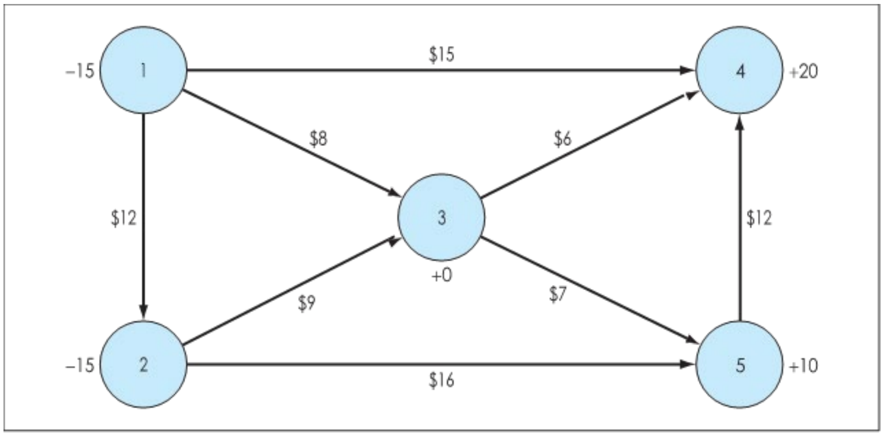
d. What is the shadow price for the Malt Resource constraint?

e. Suppose that Clayton's Brewery can acquire additional Hops at $1.40 per ounce. Should they buy more? If yes, how much? Why?

f. Suppose that on a particular day the Malt supply at Clayton's Brewery drops from 1,190 pounds to 1,000 pound, will this affect the optimal production plan and profit? If yes, what is the new plan and new profit? If no, why?

Problem 3

A used-car broker needs to transport his inventory of cars from locations 1 and 2 in Figure to used-car auctions being held at locations 4 and 5. The costs of transporting cars along each of the routes are indicated on the arcs. The trucks used to carry the cars can hold a maximum of 10 cars. Therefore, the maximum number of cars that can flow over any arc is 10.



a. Formulate an LP model to determine the least costly method of distributing the cars from locations 1 and 2 so that 20 cars will be available for sale at location 4, and 10 cars will be available for sale at location 5.

b. Use Solver to find the optimal solution to this problem.

Problem 4

Virginia Tech operates its own power generating plant. The electricity generated by this plant supplies power to the university and to local businesses and residences in the Blacksburg area. The plant burns three types of coal, which produce steam that drives the turbines that generate the electricity. The Environmental Protection Agency (EPA) requires that for each ton of coal burned, the emissions from the coal furnace smoke stacks contain no more than 2,500 parts per million (ppm) of sulfur and no more than 2.8 kilograms (kg) of coal dust. The following table summarizes the amounts of sulfur, coal dust, and steam that result from burning a ton of each type of coal.

Coal	Sulfur (in ppm)	Coal Dust (in kg)	Pounds of Steam Produced
1	1,100	1.7	24,000
2	3,500	3.2	36,000
3	1,300	2.4	28,000

The three types of coal can be mixed and burned in any combination. The resulting emission of sulfur or coal dust and the pounds of steam produced by any mixture are given as the weighted average of the values shown in the table for each type of coal. For example, if the coals are mixed to produce a blend that consists of 35% of coal 1, 40% of coal 2, and 25% of coal 3, the sulfur emission (in ppm) resulting from burning one ton of this blend is: $0.35 \times 1{,}100 + 0.40 \times 3{,}500 + 0.25 \times 1{,}300 = 2{,}110$

The manager of this facility wants to determine the blend of coal that will produce the maximum pounds of steam per ton without violating the EPA requirements.

a. Formulate an LP model for this problem.

b. Create a spreadsheet model for this problem and solve it using Solver.

c. What is the optimal solution?

d. If the furnace can burn up to 30 tons of coal per hour, what is the maximum amount of steam that can be produced per hour?

Problem 5*

A cotton grower in south Georgia produces cotton on farms in Statesboro and Brooklet, ships it to cotton gins in Claxton and Millen, where it is processed, and then sends it to distribution centers in Savannah, Perry, and Valdosta, where it is sold to customers for $60 per ton. Any surplus cotton is sold to a government warehouse in Hinesville for $25 per ton. The cost of growing and harvesting a ton of cotton at the farms in Statesboro and Brooklet is $20 and $22, respectively. There are presently 700 and 500 tons of cotton available in Statesboro and Brooklet, respectively. The cost of transporting the cotton from the farms to the gins and the government warehouse is shown in the following table:

	Claxton	Millen	Hinesville
Statesboro	$4.00	$3.00	$4.50
Brooklet	$3.5	$3.00	$3.50

The gin in Claxton has the capacity to process 700 tons of cotton at a cost of $10 per ton. The gin in Millen can process 600 tons at a cost of $11 per ton. Each gin must use at least one half of its available capacity. The cost of shipping a ton of cotton from each gin to each distribution center is summarized in the following table:

	Savannah	Perry	Valdosta
Claxton	$10	$16	$15
Millen	$12	$18	$17

Assume that the demand for cotton in Savannah, Perry, and Valdosta is 400, 300, and 450 tons, respectively.

a. Draw a network flow model to represent this problem.

b. Implement your model in Excel and solve it.

c. What is the optimal solution?